2024-28

D-4379

Time : 3 Hours

Maximum Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from all the Sections as directed.

SECTION – A

(Objective Type Questions)

1. Select the correct answer from the choice given in each of the following : 2×10=20

(a) Mitochondria can be found in ?

(i) Prokaryotic cells

(ii) Eukaryotic cells

(iii) Both Prokaryotes and Eukaryotes

(iv) Mitochondria is not present in cells

(b) Plasma membrane is a lipid bilayer was first time reported by ?

(i) Overton

(ii) Gorter and Grendel

(iii) Robertson

(iv) Singer and Nicholson

LNMU Resources

For All PYQs Visit : lnmuresources.online

(c) Ion channels perform ?

(i) Active transport

(ii) Passive transport

(iii) ATP hydrolysis

(iv) Glucose transport

(d) Choose the correct statement about Endoplasmic Reticulum :

(i) It is double membranous structure

(ii) Its membrane has continuity with the nuclear membrane

(iii) Translation occurs within its lumen

(iv) Transcription occurs on its membrane

(e) The polypeptide to be transported by Golgi complex comes from :

(i) Nucleus in coatomer coated vesicle

(ii) Nucleus in clathrin coated vesicle

(iii) Endoplasmic Reticulum in coatomer coated vesicle

(iv) Mitochondria in Clathrin coated Vesicle

(f) The enzymes of Respiratory Chain Complex (except complex II) are present in :

(i) Matrix of mitochondria

(ii) Inner membrane of mitochondria

(iii) Intermembrane space of mitochondria

(iv) Outer membrane of mitochondria

For All PYQs Visit : Inmuresources.online

(g) Nuclear pores are guarded by :

(i) Aquaporins

(ii) Nucleoporins

(iii) Voltage-dependent anion channels

(iv) Translocon apparatus

(h) Centromere remains tightly packed and generally do not code for any protein :

(i) It is constitutive heterochromatin

(ii) It is facultative heterochromatin

(iii) It is Euchromatin

(iv) None of the above

(i) Spindle assembly checkpoint works during :

(i) G1 to S phase transition

(ii) S to G2 phase transition

(iii) G2 phase

(iv) M-phase

(j) Which of the following statements is correct about meiosis :

(i) It does not have metaphase, instead it has pachytene stage

(ii) It does not have anaphase, instead it has diplotene stage

(iii) It occurs before the formation of sperm

(iv) It results in diploidy

LNMU Resources

For All PYQs Visit : lnmuresources.online

SECTION – B

(Short Answer type Questions)

2. Describe any four of the following : 5×4=20

(a) Signal peptide

(b) Diffusion through membrane

(c) F_0-F_1 Particle

(d) Nuclear lamina

(e) Cyclins and CDKs

SECTION – C

(Long Answer type Questions)

3. Answer any three of the following : 10×3=30

(a) Explain active transport through plasma membrane.

(b) Describe the structure of Mitochondria and its components.

(c) Explain the mechanism of protein sorting. Write a brief note in lysosomal targeting.

(d) Discuss the structure and parts of chromosome.

(e) Explain the events that occur in Mitosis.

LNMU Resources

For All PYQs Visit : lnmuresources.online